

Practice Quiz 08C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 08

Find and repair a gap in a proof about relations.

1. A student claims that every symmetric, transitive relation R on a set A is reflexive, and argues:

Let $a \in A$. Choose $b \in A$ with aRb . Symmetry gives bRa . Transitivity then gives aRa .

- (a) Identify the unsupported step. Verify that $R = \{(1, 1)\}$ on $A = \{1, 2\}$ is a counterexample by checking symmetry, transitivity, and failure of reflexivity.

Solution The unsupported step is choosing b with aRb : such a b may not exist. For example, take $R = \{(1, 1)\}$ on $A = \{1, 2\}$. It is symmetric because reversing its only pair changes nothing. It is transitive because the only possible two-step chain is $1R1$ and $1R1$, whose required conclusion $1R1$ holds. It is not reflexive on A , since $(2, 2) \notin R$.

- (b) Prove that the claim becomes true if one also assumes $\forall a \in A \exists b \in A (aRb)$.

Solution Suppose R is symmetric and transitive and satisfies the added hypothesis. Let $a \in A$ be arbitrary. The new hypothesis supplies $b \in A$ with aRb . By symmetry, bRa ; applying transitivity to aRb and bRa gives aRa . Since this holds for every $a \in A$, R is reflexive on A . ■